Logarit

1 Lũy thừa với số thực

$a^x \cdot a^y = a^{x+y}$ VD: $a^3 \cdot a^5 = a^8$

$(a^x)^y = a^{x \cdot y}$ VD. $(a^3)^5 = a^{15}$

$\frac{a^x}{a^y} = a^{x-y}$.

VD: $\frac{a^5}{a^3} = a^2$

$a^x \cdot b^x = (a.b)^x$ $a^3 \cdot b^3 = (a.b)^3$

$\frac{a^x}{b^x} = \left(\frac{a}{b}\right)^x$ $\frac{a^3}{b^3} = \left(\frac{a}{b}\right)^3$

$\frac{a^3}{a^5} = a^{-2} = \frac{1}{a^2}$

2. Logarit

a) Khái niệm:

$a^x = b$ Đk: $\log_a b$

$\Rightarrow x = \log_a b$ $a \neq 0$, $b > 0$.

Chú ý: +) không có logarit âm

+) " logarit cơ số 0

b) Tính chất

$\log_a 1 = 0$

$\log_a a = 1$

$a^{\log_a b} = b$

$\log_a a^b = b$

Quy tắc

$\log_a (b.c) = \log_a b + \log_a c$

$\log_a \left(\frac{b}{c}\right) = \log_a b - \log_a c$

$\log_a (b^c) = c \log_a b$

$\log_a \sqrt[n]{b} = \frac{1}{n} \cdot \log_a b$

Đổi cơ số

$\log_a b = \frac{\log_c b}{\log_c a}$

$\log_a b = \frac{1}{\log_b a}$ $(a > 0, b \neq 1)$

3. Phương trình mũ và pt logarit

Pt mũ

$a > 0,\ a \neq 1$

$a^{f(x)} = b \Leftrightarrow f(x) = \log_a b$

$a^{f(x)} = a^{g(x)} \Leftrightarrow f(x) = g(x)$

Pt logarit

$a > 0,\ a \neq 1.$

$\log_a f(x) = b \Leftrightarrow f(x) = a^b$

$\log_a f(x) = \log_a g(x) \Leftrightarrow \begin{cases} f(x) = g(x) \\ f(x) > 0 \text{ hoặc } g(x) > 0 \end{cases}$

4. Bất pt mũ và logarit

Bpt Mũ

$a > 0,\ a \neq 1$

- $a^{f(x)} > b$

$\begin{cases} b > 0,\ a > 1 \Rightarrow f(x) > \log_a b \\ b > 0,\ 0 < a < 1 \Rightarrow f(x) < \log_a b \end{cases}$

- $a^{f(x)} > a^{g(x)}$

$\begin{cases} a > 1 \Rightarrow f(x) > g(x) \\ 0 < a < 1 \Rightarrow f(x) < g(x) \end{cases}$

Bpt logarit

- $\log_a f(x) > b$

$\begin{cases} a > 1 \Rightarrow f(x) > a^b \\ 0 < a < 1 \Rightarrow f(x) < a^b \end{cases}$

- $\log_a f(x) > \log_a g(x)$

$\begin{cases} a > 1 \Rightarrow f(x) > g(x) > 0 \\ 0 < a < 1 \Rightarrow 0 < f(x) < g(x) \end{cases}$

$\sqrt{a} = a^{\frac{1}{2}}$

$\sqrt[3]{a} = a^{\frac{1}{3}}$

$\boxed{b^{\frac{2}{3}} = a} \Rightarrow b = a^{\frac{3}{2}}$

Câu 3. Cho $b > 0$ và $b^{\frac{2}{3}} = a$. Viết $b^2; \sqrt{a} \cdot b; \frac{a^6}{b^3}$ theo luỹ thừa cơ số a.

$b^2 = (a^{\frac{3}{2}})^2 = a^{\frac{3}{2} \cdot 2} = a^3$

$\sqrt{a} \cdot b = a^{\frac{1}{2}} \cdot a^{\frac{3}{2}} = a^{\frac{1}{2} + \frac{3}{2}} = a^2$

$\frac{a^6}{b^3} = \frac{a^6}{(a^{\frac{3}{2}})^3} = \frac{a^6}{a^{\frac{9}{2}}} = a^{6 - \frac{9}{2}} = a^{\frac{3}{2}}$

Câu 4. Cho $a > 0, a \neq 1$ và $a^{\frac{1}{2}} = b$. Tính:

a) $\log_a b$; $= \log_a a^{\frac{1}{2}} = \frac{1}{2}$

b) $\log_a (a^3 b^2)$; $= \log_a a^3 + \log_a b^2 = 3 + \log_a (a^{\frac{1}{2}})^2$

$= 3 + \log_a a = 3 + 1 = 4$

c) $\log_{\sqrt{a}} \left(\frac{a}{b}\right)$; $= \log_{\sqrt{a}} a - \log_{\sqrt{a}} b$

$= \log_{a^{\frac{1}{2}}} a - \log_{a^{\frac{1}{2}}} a^{\frac{1}{2}} = 2 - 1 = 1.$

d) $\log_{ab}(a\sqrt{b})$.

$+ \boxed{\log_{a^n} b = \frac{1}{n} \log_a b}$ $+ \boxed{\log_a b^n = n \cdot \log_a b}$

$b^{\frac{2}{3}} = a$

$b = a^{\frac{3}{2}}$

$\log_{a^{\frac{1}{2}}} a = \frac{1}{\frac{1}{2}} \cdot \log_a a = 2$

$\log_{a^{\frac{1}{2}}} \cdot a^{\frac{1}{2}} = \frac{1}{\frac{1}{2}} \cdot \frac{1}{2} \cdot \log_a a = 1$

$b = a^{\frac{1}{2}}$

d) $\log_{ab}(a\sqrt{b})$

$= \log_{a \cdot a^{\frac{1}{2}}} (a\sqrt{a^{\frac{1}{2}}})$

$= \log_{a^{\frac{3}{2}}} (a^{\frac{5}{4}})$

$= \frac{1}{\frac{3}{2}} \cdot \frac{5}{4} \cdot \log_a a = \frac{5}{6}$

$a \cdot \sqrt{a^{\frac{1}{2}}}$

$= a \cdot (a^{\frac{1}{2}})^{\frac{1}{2}}$

$= a \cdot a^{\frac{1}{4}}$

$= a^{1 + \frac{1}{4}} = a^{\frac{5}{4}}$

Câu 5. Giải mỗi phương trình sau:

a) $0,5^{2x^2+x-1} = \frac{1}{4}$

b) $2^{x^2-6x-\frac{5}{2}} = 16\sqrt{2}$;

c) $27^{x^2-4x+4} = 9^{x^2-4}$;

d) $0,05^{x-3} = (2\sqrt{5})^x$;

e) $\log_3 3(x-2) = -1$;

g) $\log_5 (x^2+1) + \log_{0,2}(4 - 5x - x^2) = 0$.

$a^x = b \Rightarrow x = \log_a b$

$x^2 - 6x - \frac{5}{2} = \log_2 16\sqrt{2}$

Câu 6. Giải mỗi bất phương trình sau:

[redacted] **b)** $2^{x^2-6x-\frac{5}{2}} = 16\sqrt{2}$;

a) $2^{5x+1} > 0,25$;

b) $\left(\frac{4}{9}\right)^{x-1} \leq \left(\frac{3}{2}\right)^{x+2}$;

c) $\log_{16}(3x+4) < -\frac{1}{4}$

d) $\log_{0,2}(x^2 - 6x + 9) \geq \log_{0,2}(x-3)$.

d) $0,05^{x-3} = (2\sqrt{5})^x$;

e) $\log_3 3(x-2) = -1$;

g) $\log_5 (x^2+1) + \log_{0,2}(4 - 5x - x^2) = 0$.

$\left(\frac{1}{20}\right)^{x-3} = (2\sqrt{5})^x$

$(20^{-1})^{x-3} = (2\sqrt{5})^x$

$(20)^{3-x} = (2\sqrt{5})^x$

$((2\sqrt{5})^2)^{3-x} = (2\sqrt{5})^x$

$(2\sqrt{5})^{6-2x} = (2\sqrt{5})^x$

$\Rightarrow 6 - 2x = x$

$\Rightarrow x = 2.$

$\log_a b \cdot c = \log_a b + \log_a c$

e) $\log_3 3 + \log_3 (x-2) = -1.$

ĐK: $x - 2 > 0$

$x > 2$

$\Leftrightarrow 1 + \log_3 (x-2) = -1$

$\Leftrightarrow \log_3 (x-2) = -2$

$\boxed{\log_a x = c \Rightarrow x = a^c}$

$\Leftrightarrow x - 2 = 3^{-2}$

$x = \frac{19}{9}$

g) $\log_5\left(x^2+1\right)+\log_{0,2}\left(4-5x-x^2\right)=0$.

f) k · $4-5x-x^2>0$

$\frac{-5-\sqrt{41}}{2} < x < \frac{-5+\sqrt{41}}{2}$

$-5,7 < x < 0,7$

$\log_5(x^2+1) + \log_{\frac{1}{5}}(4-5x-x^2) = 0.$

$\frac{1}{5} = 5^{-1}$

$\log_5(x^2+1) + \log_{5^{-1}}(4-5x-x^2) = 0$

$\log_5(x^2+1) - \log_5(4-5x-x^2) = 0$

$\log_5(x^2+1) = \log_5(4-5x-x^2)$

$x^2+1 = 4-5x-x^2$

$2x^2+5x-3=0 \Rightarrow \left[\begin{array}{l} x=\frac{1}{2} \text{ (tlm)} \\ x=-3 \text{ (tlm)} \end{array}\right.$